Repo report: NetworkedControlSystemOmnet-

Project folder

myplantcontroller3 is a full OMNeT++ project. Keep the .cc, .h, .ned, .msg, .xml, .ini files and makefrag. The out folder, Makefile, three .exe files, ControllerPacket_m.* and PlantPacket_m.* are build output, and result files should not be committed.

Target structure

```
repo/
  README.md  LICENSE  .gitignore
  paper/paper_TJEE_2021.pdf
  myplantcontroller3/
    Controller.cc .h .ned
    Plant.cc .h .ned
    MyPlantNode.ned  MyNetwork.ned
    ControllerPacket.msg  PlantPacket.msg
    newRoute.xml  omnetpp.ini  makefrag
    .project .cproject .oppbuildspec (hidden, keep)
```

What the code does today

- The plant updates when a control packet arrives and sends both sensor values every 10 ms.
- The controller runs a PI step, the BFDF filter and the four correlations on every packet it receives, and records them as signals.
- Between 250 s and 500 s the plant calls DamagedPlant(), which shifts sensor 1 by -5000 and sensor 2 by -2000. So the committed project runs a fault scenario only.
- The attack branches in Plant::sendPacket and Controller::sendPacket are identical, so no false data injection is active.
- Both attacker hosts have stopTime = 0 s, so no DoS runs.
- The correlations use an accumulated residual (factor 0.99), which the paper does not describe.

Minimum steps (no plots needed)

- 1. Clone the repo and copy myplantcontroller3 into it, with hidden files. Do not rename it.
- 2. Delete 55.pdf. It is a two-page Persian excerpt from the thesis, not the paper.
- 3. Put the English paper in paper/ as paper_TJEE_2021.pdf if the journal terms allow. Otherwise rely on the journal link in the README.
- 4. Add README.md, LICENSE and .gitignore at the root.
- 5. Fill in your OMNeT++ and INET versions in the README.
- 6. Run git status. You should not see out/, .exe, Makefile, _m.cc or _m.h files. If Git tracks them, use git rm -r --cached.
- 7. Commit and push.

Later, when you have time

- Rename the MyUdp comments in Plant.h and Plant.cc.
- Store old z1 and z2 before updating them in Plant_Model.
- Put attack code in the identical if/else branches of sendPacket, or remove them.
- Add [Config] sections to omnetpp.ini for each scenario.
- Add result plots to the README.

License

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